

Scalability & Future Plan

Date	3 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

Current Scalability

The application is designed to be stateless and horizontally scalable:

Attribute	Current Design
State storage	Browser session only — no server-side database required
Model serving	PKL files loaded into memory at startup — instant inference
Web server	Gunicorn WSGI — supports multiple worker processes
Deployment	Render auto-scales web workers on traffic spikes
CI/CD	GitHub push triggers Render auto-redeploy — zero-downtime updates

Future Enhancements

Phase 2: Model Improvements

Enhancement	Description	Priority
Hyperparameter tuning	GridSearchCV/RandomizedSearch for Random Forest (n_estimators, max_depth, min_samples_split)	High
Ensemble stacking	Stack LR + RF + XGBoost predictions for higher accuracy	Medium
SHAP explanations	Add SHAP feature importance to explain why each applicant was approved/rejected	High
Periodic retraining	Automate model retraining on updated credit data quarterly	Medium

Phase 3: Infrastructure Upgrades

Enhancement	Description	Priority
Database integration	Replace session storage with PostgreSQL to persist prediction history across sessions	High
User authentication	Add login system so analysts can maintain individual prediction histories	Medium
IBM Watson ML	Migrate model serving from PKL files to IBM Watson ML API for enterprise-grade SLA	Low
Docker containerization	Package app in Docker for platform-independent deployment	Medium

Phase 4: Feature Additions

Enhancement	Description	Priority
Batch prediction	Upload CSV of multiple applicants and get bulk predictions	High
Credit bureau API	Integrate with CIBIL/Experian API to fetch real credit scores automatically	Low
Audit logging	Log all predictions with timestamp and analyst ID for compliance	Medium
PDF report generation	Generate downloadable PDF report for each prediction	Low

